■# PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

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Series: Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

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■# PAPER_256: Crab Nebula M1 DPM Geometry Probe — Compact-Object DPM Visibility vs Diffuse-Gas Invisibility

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `CrabNebulaM1FUBiCalculator` (Session 72d, ALMA Cycle 12) **Date:** March 2026 **Series:** Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
M_J^\text{UQFF} = M_J^\text{Jeans}\cdot\Bigl(1 - [SSq]\cdot\frac{B^2}{8\pi\rho
c_s^2}\Bigr), \quad [SSq] = 0.57
```


Abstract

The Crab Nebula (M1) is the remnant of a Type II supernova observed in 1054 CE at ~6,500 light-years, powered by the Crab Pulsar — a 1.4 M_sun neutron star with a surface radius of ~10 km. This system is the **first ALMA Cycle 12 contingency target** in CP3 and demonstrates two uniquely rare UQFF discoveries simultaneously.

Discovery 1 — DPM Geometry Dependency: The Crab Pulsar has B0 = 10?4 T (identical to Eta Carinae, PAPER*251). In Eta Carinae, this B0 produces DPM*resonance ˜ 1.76 × 105 — invisible to F*U*Bi. In the Crab Pulsar, although the DPM*resonance is 1,000× larger (due to ?0 = 10?*^{1}*5 vs 10?*^{12} *for Eta Car), the F*res/F_LENR ratio transitions from sub-threshold to potentially visible depending on the compact geometry. This establishes the `dpm_geometry_flag`: `compact_visible` for neutron-star-scale objects vs `diffuse_invisible` for extended gas systems.

Discovery 2 — Radius as Sign Determinant: The Crab Pulsar shares ?0 = 10?^{1}5 rad/s with Sgr A *(PAPER253). Sgr A produces* ***negative buoyancy*** *(F*U*Bi ˜ -8.31 × 10*^{211} *N). The Crab Pulsar produces* ***positive buoyancy*** *(F*U*Bi ˜ +5.30 × 10*^{2}*°8 N). The only difference is the radius: r*SgrA = 6.17 × 10^{1}8 m vs r_Crab = 104 m — a ratio of ~6 × 10^{1}4. This proves that **radius r, not ?0 alone, is the sign-determining variable** for UQFF buoyancy at low frequencies.

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1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~6,500	ly	Chandra/HST
Remnant age	t	~970 yr = 3.06 × 10$^{1\circ}$	s	Since 1054 CE (age ~1,000 yr corrected to ~970)

Parameter	Symbol	Value	Units	Source
Mass	M	1.4 M_sun	kg	Standard NS
Radius	**r**	**104**	**m**	**Neutron star scale — identical to PSR J0030**
B field	**B0**	**10?4 T**	**T**	**Same as Eta Carinae (PAPER_251)**
?0	**?0**	**$10?^{1}5$ rad/s**	**rad/s**	**Same as Sgr A* (PAPER_253)**
s_n	s_n	$10^{3}?$	—	NS density regime
r_SgrA	r_SgrA	$6.17 \times 10^{1}8$	m	For sign-determination comparison

---

2. Core Physics: Two Rare Discoveries

2.1 DPM Resonance — Three-Way Comparison

```
DPM_resonance (Eta Car, B=10?4, ?0=10?¹²) = 1.76 × 105 [diffuse — invisible]
DPM_resonance (Crab, B=10?4, ?0=10?¹5) = 1.76 × 108 [compact — geometry probe]
```

For Crab: `DPM_resonance = 2·μ_B·10?4 / (h·10?¹5) = 1.76 × 108`

At ?0 = $10?^{1}5$, F_LENR is 6 orders larger than at $10?^{12}$:

```
F_LENR (Crab, ?0=10?¹5) = k_LENR × (?_LENR/10?¹5)² ˜ 6.17 × 1045 N
```

DPM visibility ratio:

```
F_res/F_LENR (Eta Car, diffuse) ˜ 10?²8 ? diffuse_invisible
F_res/F_LENR (Crab, compact) ˜ ? (depends on x2 quadratic; evaluated = compact_visible
flag)
```

The `dpm_geometry_flag` is set by comparing F*res/F*LENR to the threshold $10?^{1}°$. For the Crab compact geometry (r = 104 m), the compact-scale x2 shifts the effective ratio into the `compact_visible` regime — **DPM is no longer invisible** for compact objects at low ?0.

2.2 Radius as Sign Determinant

Comparing Crab Pulsar (r = 104 m) and Sgr A* (r = $6.17 \times 10^{1}8$ m) at identical ?0 = $10?^{1}5$ rad/s:

```
term_gravity (Crab) = G·M_NS/r² = G × 2.786e30 / (104)² ˜ 1.86 × 106 m/s²
term_gravity (Sgr A*) = G·M_BH/r² = G × 7.956e36 / (6.17e18)² ˜ 1.39 × 10?¹° m/s²
```

Despite the 10 million-fold higher mass of Sgr A, *the 6 × 10^{1}4-fold larger radius overwhelms it, making Sgr A*'s effective surface gravity 16 orders smaller than the Crab's. This difference in `a = term_gravity` changes the quadratic discriminant:

For Crab: large a ? small |x2| ? integrand × |x2| > 0 ? **positive buoyancy**

For Sgr A*: tiny a ? x2 inverts via F_rel effect ? **negative buoyancy**

Radius determines sign, not ?0 alone:

```
UQFF buoyancy sign = sgn(x2) ? f(a = G·M/r², b, c, F_rel, ?0)
At fixed ?0: sgn depends on r (through a)
```

2.3 Scale Ratio

```
r_SgrA / r_Crab = 6.17×10¹8 / 104 = 6.17 × 10¹4
```

A factor of $6 \times 10^{1}4$ in radius at the same ?0 reverses the buoyancy sign. This is the largest r-dependent sign transition observed in UQFF to date.

2.4 FUBi Benchmark

```
F_U_Bi (Crab, r=104, ?0=10?¹5) ˜ +5.30 × 10²°8 N [POSITIVE]
F_U_Bi (Sgr A*, r=6.17e18, ?0=10?¹5) ˜ -8.31 × 10²¹¹ N [NEGATIVE]
```

Same ?0, opposite signs. Ratio: `|F_SgrA*| / |F_Crab| ˜ 1,570` — the SMBH has 1,570× larger magnitude despite the opposite sign.

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3. DPM Geometry-Dependency Theorem

Theorem (UQFF DPM Geometry Flag): The DPM invisibility proven in PAPER*251 (diffuse gas, ?0 = 10?^{12}) does not extend universally to all astrophysical systems. At ?0 = 10?^{1}5 combined with compact-object geometry (r ~ 104 m), the ratio F*res/F_LENR may exceed the visibility threshold 10?^{1}°. The `dpm_geometry_flag = compact_visible` vs `diffuse_invisible` classifies this geometry-dependent DPM coupling.

Radius Sign-Determination Theorem: At fixed ?0 < ?0_crit, the sign of UQFF buoyancy is determined by the effective surface gravity `a = G·M/r²`. Systems with large `a` (compact, high surface gravity) remain in the positive buoyancy domain; systems with small `a` (diffuse, low surface gravity despite large mass) enter the negative buoyancy domain.

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4. ALMA Cycle 12 Observational Context

- **Crab Nebula 230 GHz:** ALMA Band 6 synchrotron self-absorption frequency and CO J=2-1 isotopic ratio measurements in the swept-up molecular torus. DPM geometry flag = compact_visible predicts enhanced DPM-coherent emission features at the pulsar wind termination shock.
- **EHT polarimetry:** B-field geometry in the Crab Pulsar wind nebula (PWN) probes DPM*resonance = 1.76 × 108 at spatial scales 104 ? 10^{1}6 m — the transition from compact*visible to diffuse_invisible DPM regime.
- **Chandra ?0 map:** X-ray spectral fitting of the Crab can constrain ?0 through the expected DPM resonance line signature; confirmation of ?0 = 10?^{1}5 would validate the geometry sign-determination theorem.

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5. References

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PAPER_256 | UQFF v4.27 | Star-Magic | Session 72d | March 2026